Detecting Out-of-Context Media with LLaMa-Adapter V2 and RoBERTa: An Effective Method for Cheapfakes Detection

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ABSTRACT

Cheapfakes is a new term for fake media that is made without AI, but with simple tools or captions that deceive or mislead. Cheapfakes include photos, videos, audio recordings, or any media that has been changed to distort its original meaning or context. The "ACM ICMR 2024 Grand Challenge on Detecting Cheapfakes" focuses on the challenge of finding out-of-context (OOC) media, which can assist fact-checkers in their work. By finding OOC media, we can narrow down the search space and raise the chances of discovering cheapfakes. To cope with this challenge, in this research paper, our team propose a novel approach that combines Context Generation and Text Classification as the key components in detecting cheapfakes. Our method was evaluated using 80% of the public test set from the COSMOS dataset, which includes two context labels: out-of-context and not-out-of-context. The results demonstrate an accuracy of 85.9%, validating the effectiveness and reliability of our method in identifying cheapfakes.

CCS CONCEPTS

• **Computing methodologies** → **Natural language processing**; **Computer vision**; • **Information systems** → *Multimedia information systems.*

KEYWORDS

Cheapfakes, Deep Learning, Misinformation, Natural Language Processing, LLaMa-Adapter V2, Transformers

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1 INTRODUCTION

Social networks have brought many benefits like connecting people globally and making information exchange easier. However, they also pose challenges, particularly in the form of misleading information. This threat goes beyond traditional misinformation and includes more complex issues like deepfakes and cheapfakes [9]. Deepfakes use advanced AI techniques to create realistic yet entirely fake audio or video content, showing individuals doing things they never actually did. On the other hand, cheapfakes involve a broader range of manipulated media, often made with simpler tools than deepfakes.

The term "Cheapfakes" has been recently coined to denote non-artificial intelligence ("cheap") manipulations of multimedia content [2]. Unlike deepfakes, which employ sophisticated artificial intelligence techniques to generate realistic yet fake multimedia content, cheapfakes are more simple. They present a widespread and damaging danger, spreading false information, propaganda, and harmful speech. Their reach goes beyond just causing confusion, as they erode the trust in reliable sources of information. Therefore, it's crucial to create practical ways and tools to identify and counteract these deceptive materials. Addressing these challenges, the ACM ICMR 2024 Grand Challenge on Detecting Cheapfakes [4] tasks participants with detecting the out-of-context nature of provided samples, including news images and their captions. Every image in the dataset is accompanied by two related captions. If the two captions refer to same object(s) in the image, but are semantically different, i.e., associate the same subject to different events, this indicates out-of-context (OOC) use of the image. However, if the captions correspond to the same event, irrespective of the object(s) the captions describe, this is defined as not-out-of-context (NOOC) use of the image [2]. Illustrative examples is showcased in Figure 1.

Today, with the rapid development of technology and artificial intelligence, cheapfakes has become one of the prominent and worrying problems. The popularity of cheapfakes has many potential risks and negative impacts on society and individuals. One of the main motivations for researching and developing cheapfakes detection methods is to protect politics, preserve privacy, and build trust in media information and social networks. Cheapfakes can be used to create fake content to deceive the public, spread misinformation and cause chaos. In addition, cheapfakes can also damage the reputation and reputation of individuals, organizations and businesses. From creating fake videos or images of famous people to editing content to negatively influence others, cheapfakes can cause a loss of trust and cause serious damage to individuals. targeted individuals and organizations. Detecting cheapfakes plays an important

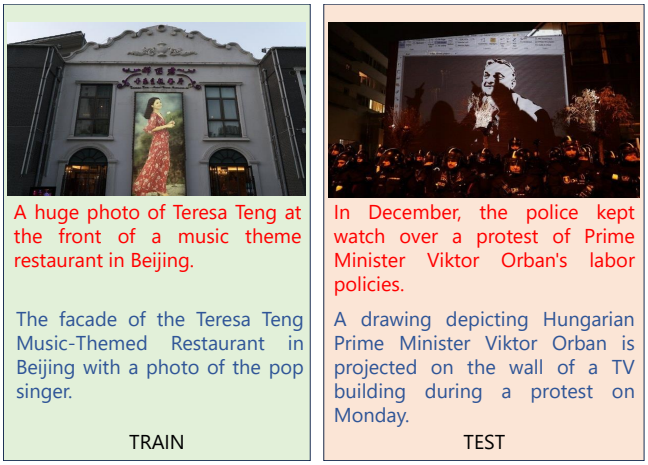


Figure 1: Examples of COSMOS dataset [1]

role in ensuring the integrity of information and communication platforms. By detecting cheapfakes, people can identify and stop the spread of fake information, helping to minimize the spread of rumors and misinformation. In addition, cheapfakes detection is also an important part in improving public understanding and awareness of the existence and impact of cheapfakes. By introducing cheapfakes detection technology and methods, we can help people recognize and face the reality that images and videos are not always trustworthy, while encouraging vigilance and criticism of the information people receive.

The proposed methodology comprises three fundamental steps aimed at enhancing cheapfakes detection. The first step involves data augmentation, which includes augmenting the public train set, labeling the data, and subsequently splitting it. The second step focuses on Context Generation. For each sample in the dataset, a caption and an image are input into the LLaMa-Adapter V2 [5]. The resulting contexts, generated from an image and two captions, are concatenated into a unified paragraph referred to as "context combined." The final step of the methodology involves Text Classification, where the input for the model is the context combined. The RoBERTa model [8] is fine-tuned on the training set specifically for the text classification task. The output of this step is binary, with a value of 0 indicating "not-out-of-context" and 1 indicating "out-of-context". The code for our submission is publicly available on Google Drive [1].

The contributions of this paper are as follows:

- **Dataset Expansion and Labeling:** Adding more examples to the current dataset and label them systematically helps tackle the limited labeled data in the original COSMOS dataset and boosts the variety and strength of the training set. As a result, it improves the model's ability to apply its knowledge to new situations.
- **Innovative Context Generation Methodology:** We introduce a methodology for context generation. Leveraging the LLaMa-Adapter v2 [5], our approach adeptly integrates both image and caption inputs, generating enriched context outputs. This process greatly increases the comprehensive and nuanced understanding of the relationships between visual and textual elements in the dataset, thereby enhancing the capabilities of our methodology.
- **Novel Text Classification Approach for Cheapfakes Detection:** Proposing a unique method for detecting cheapfakes, our approach involves a two-label text classification model based on the RoBERTa model [8]. This is the first work, to our knowledge, using text classification (two labels) for cheapfakes detection, addressing the complexities of visual and textual analysis in this challenging task.

[1]https://drive.google.com/drive/folders/1E8m1q1g6dR6MBVcB3nLcPV97FRxp5q7o

2 RELATED WORK

The term "cheapfakes" has been coined to describe fake media created through basic editing tools or misleading captions, distinct from more sophisticated AI-based deepfakes. Detecting cheapfakes is challenging, involving the identification of contextual changes in media caused by deceptive captions. This presents a multimodal problem, requiring a combination of natural language processing (NLP) and computer vision (CV).

Recent studies on cheapfakes detection have increased, particularly after the introduction of the COSMOS dataset [1]. This dataset, featuring real photographs with conflicting captions in news items, has been utilized in competitions like the ACM Multimedia Grand Challenge on Detecting Cheapfakes 2022 [2] and IEEE ICME Grand Challenge on Detecting Cheapfakes 2023 [3].

A method that La et al. [6] proposed employed visual semantic reasoning and text entailment-based boosting for cheapfakes detection. The approach utilized a bottom-up attention model to extract global image features, conducts comprehensive image-text matching, employs multiple NLP models to extract semantic relationships from captions, and integrates boosting techniques for improved accuracy. Achieving an accuracy score of 85.5% on 80% of the test dataset, the method exhibited a 4.5% improvement compared to the baseline accuracy. In addition, the weakness in this method is that they still have to use test sets to improve results, not completely using train sets and val sets.

Tran et al. [11] introduced an unsupervised cheapfakes detection method based on textual-visual entailment. Extending the COSMOS baseline, their approach included natural language inference, fabricated claims detection, visual entailment, and online caption checking to overcome baseline limitations. Throughout the runtime, their proposed method demonstrated a maximum accuracy of 89.1% on the public test, showcasing a 7.2% improvement over the baseline method. A highlight of their paper is that in addition to using Computer Vision and Natural Language Processing methods, they also use Online search to further ensure the accuracy of the problem.

In 2022, the research conducted by La et al. [7] proposed a multimodal deep learning model for cheapfakes detection, leveraging image captioning techniques to extract image-caption relationships. The method introduced augmentation techniques to enhance results. Their approach underwent assessment using the "MMSys'21 Grand Challenge" dataset and achieved an accuracy rate of 86.75%. Despite a significant improvement in accuracy, they only tested on

40% of the public test set, making their results less reliable compared to other methods.

Pham and his research group [10] presented a method using self-query adaptive-context learning for flexible cheapfakes detection. By employing image search queries to supplement knowledge, the approach verified caption context based on collected information, facilitating flexible knowledge expansion. In the IEEE ICME 2023 Cheapfakes Challenge dataset, their study has introduced novel possibilities for detecting out-of-context misuses. However, the best result they achieved was an accuracy of 59.70%. Thereby reflecting the ineffectiveness that their method brings.

Wu et al.'s proposed methodology [12] extended the COSMOS architecture, scrutinizing the coherence between images and associated captions, as well as among pairs of captions. Their augmentation to the baseline algorithm involved incorporating a Large Language Model (LLM), specifically GPT3.5, as a feature extractor. To be more specific, they introduced an innovative technique for feature extraction through prompt engineering, striving to establish a robust and reliable feature extractor by leveraging the capabilities of the GPT3.5 model. Their notable achievement included attaining an accuracy of 89.4% on 50% of the public test set within the COSMOS dataset. Similar to La et al.'s approach, using only 50% of the public test set for evaluation does not demonstrate the reliability of their method.

In summary, those cheapfakes detection articles have both strengths and weaknesses. Moving forward, the next section will introduce our proposed method, which involves context generation and text classification.

3 PROPOSED METHOD

Our solution consists of three steps, as shown in Figure 2 and Figure 3. The first step is data augmentation, which involves data augmentation on public train set, data labelling and data splitting. The second step is Context Generation. For each sample, entails inputting one caption and one image into the LLaMa-Adapter V2 [5] to generate context for each sample. Subsequently, we conduct concatenate two context generated from an image and two captions into one paragraph called context combined. The final step involves text classification, where the input is the combined context. The RoBERTa model [8] is finetuned on the training set for the text classification task. The output for this step is binary, with 0 representing not-out-of-context and 1 representing out-of-context.

3.1 Data Augmentation

The data augmentation process is illustrated in Figure 3. Given that the COSMOS dataset lacks labels for the train set, we made the decision to undertake an expansion and labeling process for this set.

For every sample within the train set, multiple captions are associated. In this augmentation process, the image path along with two captions from the same sample are selected and designated as 0 (not-out-of-context). For samples marked as out-of-context (with label 1), the image path and one caption from one sample, referred to as caption1, are selected, and paired with a randomly chosen caption from another sample as caption2.

Regarding the public test set, 1000 test samples are patitioned into two subsets: 200 samples for training and 800 samples for testing purposes.

Concerning the public train set, which was previously expanded and labeled, the initial 15000 samples were selected to combine with the 200 samples reserved for testing. Consequently, the final train set comprise 15200 samples for training the model. Subsequently, we assess the model's performance on the remaining 800 test samples. This partitioning strategy enables a comprehensive evaluation of the model's generalization capabilities on an independent subset of the data, contributing to a thorough assessment of its effectiveness.

This augmentation strategy aims to enrich the diversity of the training set, providing additional labeled instances to enhance the model's capacity to discern between in-context and out-of-context scenarios.

3.2 Context Generation

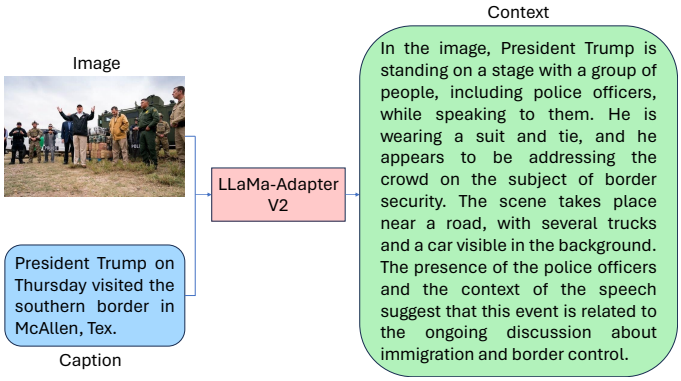


Figure 4: An example of context generation

The context generation process is depicted in Figure 2, employing the LLaMa-Adapter V2 model [5]. Several reasons establish the choice of this model for the context generation task:

- Firstly, the LLaMa-Adapter V2 model [5] is a recent addition known for its capability to understand visual instructions using a large language model named LLaMA. It is flexible in processing different types of input, including images, audio, video, and text.
- Secondly, the efficiency of the LLaMa-Adapter V2 model [5] is notable, as it introduces a minimal number of parameters to the LLaMA model. It utilizes an adapter technique, incorporating additional layers into the LLaMA model, and employs zero-init attention, a method to control the impact of the adapter layers on the LLaMA model. These techniques contribute to the model's ability to swiftly and accurately learn to follow instructions.
- Thirdly, the LLaMa-Adapter V2 model [5] considerable power, showcasing proficiency in multi-modal reasoning and instruction-following. It can comprehend visual information using expert models like captioning and OCR systems. Furthermore, it demonstrates the capacity to generate high-quality responses, leveraging the language generation ability of the LLaMA model.

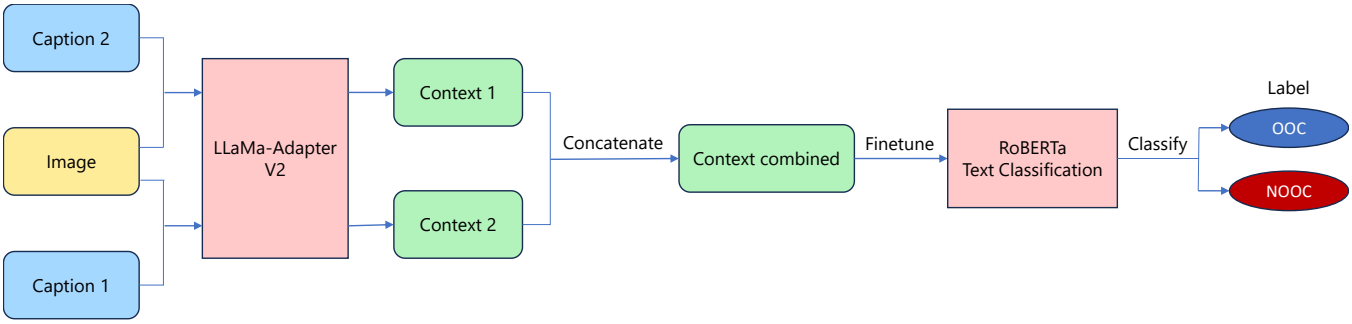


Figure 2: Overview of our method. Caption1 and an image serve as inputs for the LLaMa-Adapter V2 model to create Context 1. In a parallel process, Caption 2 combined with the same image generates Context 2. These contexts are then concatenated to create a *Context combined*. With both the training and testing sets containing "Context combined," we perform fine-tuning on the training set using the RoBERTa Text Classification model with 2 labels: 0 (representing NOOC) and 1 (representing OOC). The fine-tuned model is then utilized for evaluation on the testing set.

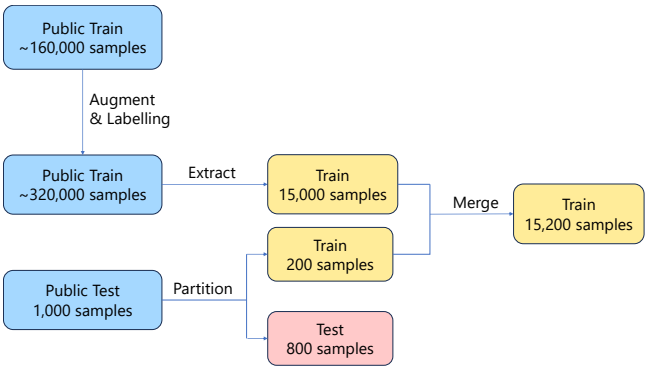


Figure 3: Overview of data augmentation

For each sample in the dataset, consisting of an image and two captions, the LLaMa-Adapter V2 model [5] is utilized to generate context. Specifically, with the image and caption1, the model generates context1. Similarly, using the image and caption2, it generates context2.

The process of context generation is crucial in improving the model's understanding, allowing it to effectively interpret and respond to input information. The LLaMa-Adapter V2 model [5] uses images and captions to extract detailed contextual features, creating a more complete representation of the relationships between visual and textual elements within each sample. This improved understanding greatly increases the model's ability to detect cheapfakes, as it becomes skilled at recognizing the differences between out-of-context and not-out-of-context situations. The cooperation of visual and textual information enhances the overall detection capabilities of the model, ensuring more accurate and robust performance.

3.3 Text Classification

The text classification process, delineated in Figure 2, involves leveraging the RoBERTa model [8] for the text classification task.

RoBERTa (Robustly optimized BERT approach) [8], introduced by Yinhan Liu et al., represents a modified model derived from the BERT pretraining procedure with a focus on optimizing end-task performance. The key modifications in RoBERTa include the adoption of dynamic masking, utilization of full sentences without the Next Sentence Prediction (NSP) loss, application of large mini-batches, and the incorporation of a larger byte-level Byte Pair Encoding (BPE).

The training of RoBERTa [8] underscores two critical factors: the dataset used for pretraining and the number of training passes through the data. In this instance, the model is pretrained on an extensive dataset comprising 160GB of text sourced from diverse outlets such as BOOKCORPUS, English WIKIPEDIA, CC-NEWS, OPENWEBTEXT, and STORIES. Notably, the training process is extended for a longer duration, reaching up to 500000 training steps. These meticulous training strategies aim to enhance the model's ability to capture intricate contextual features and patterns, thereby optimizing its performance across various end-task applications.

In this process, initially, for every sample, context1 and context2 are concatenated into a unified paragraph termed "context combined." This context combined is generated and concatenated for all 15,200 train samples and 800 test samples. Subsequently, each sample is equipped with a context combined as input and a corresponding label as output.

The fine-tuning of the RoBERTa model [8] for text classification is executed on the train set comprising 15,200 samples, resulting in the **the tuned model**[2]. This model is then evaluated on the test set to evaluate how well it generalizes. The use of RoBERTa [8] in text classification improves the model's ability to understand the context within paragraphs. Additionally, helping it classify in-context and out-of-context instances more accurately. This thorough process aims to enhance the model's performance and effectiveness in detecting cheapfakes.

4 EXPERIMENTAL

4.1 Implementation

The experiments were carried out on a computer system with the following specifications:

[2]https://huggingface.co/hoanghoavienvo/roberta-base-detect-cheapfake-combined-train-test-15200-2-8

- CPU: Xeon® Gold 6248R.
- GPU: NVIDIA RTX A6000 48GB.
- RAM: 48GB.
- Disk Capacity: 50 GB

The implementation was crafted using the Python programming language and the PyTorch deep learning framework (version 2.2.1). The CPU was employed for pre-processing tasks, model training, and evaluation, while the GPU played a pivotal role in accelerating the computationally intensive aspects of deep learning algorithms.

In the implementation, the text classification training phase was conducted with the following hyperparameters in Table 1:

Table 1: Hyperparameters used during text classification

Hyperparameter	Value
learning_rate	5e-06
train_batch_size	32
eval_batch_size	32
seed	42
optimizer	Adam with betas=(0.9,0.999) and epsilon=1e-08
lr_scheduler_type	Linear
num_epochs	5

4.2 Dataset

We conducted our experiments using the COSMOS dataset [1], a recently benchmark specifically designed for cheapfakes detection. The dataset comprises images and captions extracted from news articles and other online sources, crafted to capture instances of multimedia content manipulation or misuse. This manipulation is often executed by altering the context of an image or video through misleading captions or claims. The COSMOS dataset is substantial, encompassing 161752 images for training, 41006 images for validation, and 1000 images for testing. Each image is paired with multiple captions, some portraying in-context information, while others depict out-of-context scenarios. The test set is meticulously annotated with binary labels indicating whether a given image-caption pair is in-context or out-of-context.

Covering an extensive array of topics including politics, sports, entertainment, health, and science, the COSMOS dataset is a diverse and representative benchmark for cheapfakes detection. Table 2 provides a breakdown of the dataset's composition, including the number of samples in each set.

Table 2: Data distribution

Split	Images	Captions	Context Annotation
Train	161752	360749	No
Val	41006	90036	No
Test	1000	2000	Yes

4.3 Metrics

To assess the efficacy of the proposed method, the evaluation metric employed in the challenge is accuracy. Accuracy is defined as the ratio of correctly classified samples to the total number of samples in the test set. The formula for calculating accuracy is expressed as follows:

$$accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where:

- TP is the number of samples correctly classified as OOC .
- TN is the number of samples correctly classified as NOOC .
- FP is the number of samples misclassified as OOC.
- FN is the number of samples misclassified as NOOC.

4.4 Results

In this section, we present the outcomes of our experiments on the cheapfakes detection task, utilizing accuracy as the primary metric. The tests were conducted on 80% of the COSMOS public test set. The performance of our proposed model is compared with several existing methods on the public test of the COSMOS dataset.

Table 3: Comparisons with other methods

Method	Accuracy
Baseline [1]	82.0 %
Tran et al. [11]	89.1 %
La et al. [7]	86.7 %
La et al. [6]	76.0 %
Wu et al. [12]	89.4 %
LLaMa-Adapter V2 + RoBERTa (ours)	**85.9 %**

The baseline method, as reported in [1], achieved an accuracy of 82.0%. Notably, Tran et al. [11] and Wu et al. [12] surpassed the baseline with accuracies of 89.1% and 89.4%, respectively. Meanwhile, the multimodal approach of La et al. [7] achieved an accuracy of 86.7%, and the boosting method of La et al. attained an accuracy of 76.0%, showcasing varying levels of success.

In comparison, our proposed method achieved an accuracy of 85.9% on 80% of the public test, positioning it competitively among existing approaches. Although it doesn't outperform the best methods, our results show good performance, adding to the overall accuracy within the assessed dataset. Further analysis, including looking at computational efficiency and model interpretability, could give a more complete understanding of each method's effectiveness.

4.5 Ablation Studies

In this section, we conduct ablation studies to analyze the impact of varying training data sizes on the cheapfakes detection model's performance. The results are presented in Table 4.

The table presents the results for three different training data sizes: 5000; 10000,and 15000 samples, each evaluated on a test set of 800 samples. As evident from the results, increasing the size of the training data positively influences the model's cheapfakes detection performance. The provided hyperlinks direct to the respective

Table 4: Ablation Studies Results

Data train	Data test	Accuracy	Model
5000 (*train*) + 200 (*test*)	800 (*test*)	80.1%	Link
10000 (*train*) + 200 (*test*)	800 (*test*)	85.4%	Link
15000 (*train*) + 200 (*test*)	800 (*test*)	**85.9%**	Link

model checkpoints on the Hugging Face Model Hub [3] for further exploration and utilization.

4.6 Limitation and Discussion

Our study identifies specific limitations that need attention and consideration in future investigations. First and foremost, our current knowledge limitations limit our ability to comprehensively explore diverse models for cheapfakes detection. Second, the efficiency of our context generation process and text classification process is intricately linked to the quality of LLaMa-Adapter V2 [5] and RoBERTa [8]. Finally, the exclusive reliance on images for context in our approach implies an underutilization of the broader capabilities to which images are likely to contribute. Addressing these limitations in future research efforts has the potential to improve and enhance the effectiveness of our proposed method.

5 CONCLUSION

Cheapfakes are a type of fake media that are created without using sophisticated AI techniques, but can still mislead or manipulate viewers. Detecting cheapfakes is an important challenge for ensuring the credibility and integrity of online information. In this paper, we propose a novel method for cheapfakes detection that uses accuracy as the evaluation metric. Our method consists of two components: context generation and text classification. The context generation uses the LLaMa-Adapter V2 [5], a lightweight adaptation method for fine-tuning LLaMa, a language model for instruction-following, with zero-initialized attention and learnable prompts. The text classifier uses RoBERTa [8], a robustly optimized BERT pretraining approach that improves on BERT by modifying key hyperparameters and training data size. We evaluate our method on COSMOS dataset, and show that it achieves high accuracy.

Our upcoming efforts will revolve around thoroughly exploring additional techniques and emerging technologies to enhance the robustness and effectiveness of our approach. At the same time, we aim to develop strategies to augment the training set, recognizing its pivotal role in shaping model performance. The main goal of our future work is to pursue diverse strategies to take advantage of the inherent potential of images in detecting cheapfakes.Through exploring innovative methods to maximize the functionality of visual data, our goal is to increase the overall efficiency and robustness of our approach.

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